

Bash Script : Lab 3

1. Write a script called mycase, using the case utility to checks the type of character entered by a user: a. Upper Case. b. Lower Case. c. Number. d. Nothing

```
#!/bin/bash
shopt -s extglob
var=$1
case "$var" in
+([a-z])) echo "Lower Case." ;;
+([A-Z])) echo "Upper Case." ;;
+([0-9])) echo "Numbers." ;;
*) echo "Invalid input." ;;
esac
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh nagy
Lower Case.
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh NAGY
Upper Case.
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh 123
Upper Case.
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
```

2. Enhanced the previous script, by checking the type of string entered by a user: a. Upper Cases. b. Lower Cases. c. Numbers. d. Mix. e. Nothing.

```
#!/bin/bash
shopt -s extglob
var=$1
case "$var" in
+([a-z])) echo "Lower Case." ;;
+([A-Z])) echo "Upper Case." ;;
+([0-9])) echo "Numbers" ;;
+([a-zA-Z0-9])) echo "Mixed" ;;
esac
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh 123
Numbers
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh NAGY
Upper Case.
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh Nagy
Mixed
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh nagy
Lower Case.
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
```

3. Write a script called mychmod using for utility to give execute permission to all files and directories in your home directory.

```
#!/bin/bash
dir=$1
if [[ ! -d "$dir" ]]; then
    echo "wrong directory"
    exit 1
fi
for item in "$dir"/*
do
    chmod +x "$item"
done
echo "Done."
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh .
Done.
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
```

4. Write a script called mybackup using for utility to create a backup of only files in your home directory.

```
#!/bin/bash
dir=$1
if [[ ! -d "$dir" ]]; then
    echo "Wrong directory"
    exit 1
fi
backup_dir="$dir/mybackup"
mkdir -p "$backup_dir"
for item in "$dir"/*
do
    if [[ -f "$item" ]]; then
        cp "$item" "$backup_dir"
    fi
done
```

```
done
echo "Backup completed."
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh .
Backup completed.
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ls mybackup/
Lab3.md  script.sh
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
```

5. Write a script called mymail using for utility to send a mail to all users in the system. Note: write the mail body in a file called mtemplate.

```
#!/bin/bash
if [[ ! -f "mtemplate" ]]; then
    echo "Mail template file (mtemplate) not found."
    exit 1
fi
for user in $(cut -d: -f1 /etc/passwd)
do
    echo "Sending mail to $user"
    mail -s "System Notification" "$user" < mtemplate
done
echo "All mails sent."
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ echo "Hello World" >> mtemplate
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh
Sending mail to root
Sending mail to daemon
Sending mail to bin
Sending mail to sys
Sending mail to sync
Sending mail to games
Sending mail to man
Sending mail to lp
Sending mail to mail
Sending mail to news
```

6. Write a script called chkmail to check for new mails every 10 seconds. Note: mails are saved in /var/mail/username.

```
#!/bin/bash

mailfile="/var/mail/$USER"
if [[ ! -f "$mailfile" ]]; then
    echo "Mail file not found for $USER"
    exit 1
fi
```

```
oldcount=$(wc -l < "$mailfile")
while true
do
    sleep 10
    newcount=$(wc -l < "$mailfile")
    if [[ $newcount -gt $oldcount ]]; then
        echo "You have new mail!"
        oldcount=$newcount
    fi
done
```

8. Create the following menu:

- a. Press 1 to ls
- b. Press 2 to ls -a
- c. Press 3 to exit
- Using select utility then while utility.

```
#!/bin/bash
PS3="Enter ur Option ==> "
options=("ls" "ls -a" "Exit")
while true
do
    select op in "${options[@]}"
    do
        case $REPLY in
            1) ls; break; ;;
            2) ls -a; break; ;;
            3) exit 0; break; ;;
        esac
    done
    echo ""
done
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh
1) ls
2) ls -a
3) Exit
Enter ur Option ==> 1
Lab3.md mtemplate mybackup script.sh

1) ls
2) ls -a
3) Exit
Enter ur Option ==> 2
. .. Lab3.md mtemplate mybackup script.sh

1) ls
2) ls -a
3) Exit
Enter ur Option ==> 3
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
```

9. Write a script called myarr that ask a user how many elements he wants to enter in an array, fill the array and then print it

```
#!/bin/bash
read -p "Enter Array Size: " n
arr=()
for (( i=0; i<n; i++ ))
do
    read -p "Enter element $((i+1)): " elem
    arr+=("$elem")
done
echo "Array elements are: ${arr[@]}"
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh
How many elements do you want to enter? 3
Enter element 1: 2
Enter element 2: 3
Enter element 3: 4
The array elements are: 2 3 4
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
```

10. .Write a script called myavg that calculate average of all numbers entered by a user. Note: use arrays

```
#!/bin/bash
read -p "Enter Array Size: " n
arr=()
for (( i=0; i<n; i++ ))
do
    read -p "Enter number $((i+1)): " num
    numbers+=("$num")
    sum=$((sum + num))
done
```

```
done
avg=0
if [[ $n -gt 0 ]]; then
    avg=$((sum / n))
fi
echo "Average: $avg"
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh
Enter Array Size: 4
Enter number 1: 1
Enter number 2: 7
Enter number 3: 4
Enter number 4: 2
Average: 3
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
```

11. Write a function called mysq that calculate square if its argument

```
#!/bin/bash
read -p "Enter a number: " n
result=$((n*n))
echo "Square of $n is: $result"
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$ ./script.sh
Enter a number: 5
Square of 5 is: 25
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab3$
```